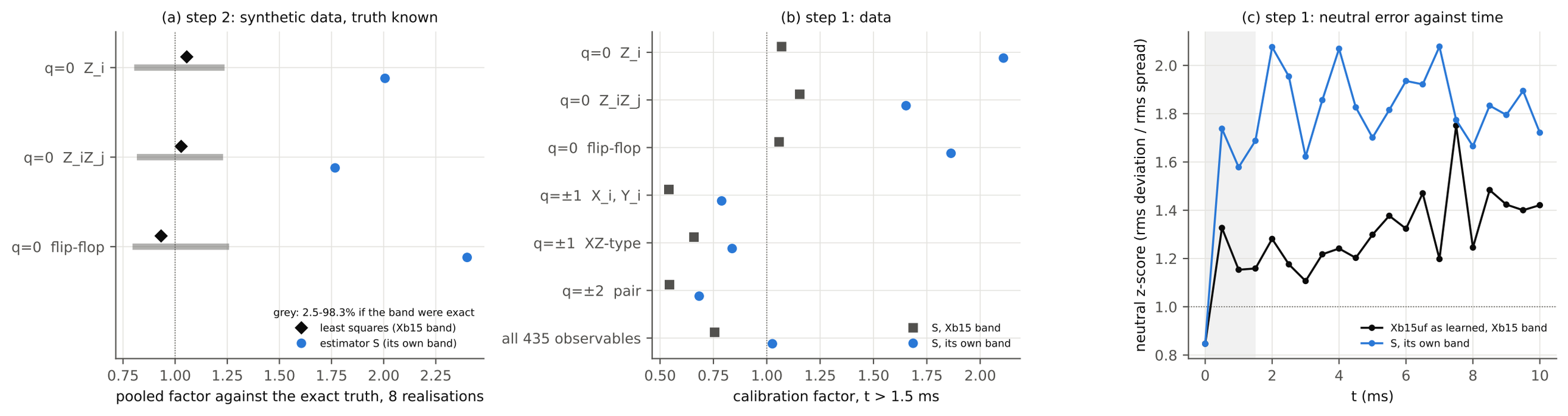
(a) step 2: synthetic data, truth known
q=0 Z_i
q=0 Z_iZ_j
q=0 flip-flop
grey: 2.5-98.3% if the band were exact
least squares (Xb15 band)
estimator S (its own band)
0.75
1.00
1.25
1.50
1.75
2.00
2.25
pooled factor against the exact truth, 8 realisations
(b) step 1: data
q=0 Z_i
q=0 Z_iZ_j
q=0 flip-flop
q=±1 X_i, Y_i
q=±1 XZ-type
q=±2 pair
all 435 observables
S, Xb15 band
S, its own band
0.50
0.75
1.00
1.25
1.50
1.75
2.00
calibration factor, t > 1.5 ms
(c) step 1: neutral error against time
neutral z-score (rms deviation / rms spread)
0.8
1.0
1.2
1.4
1.6
1.8
2.0
Xb15uf as learned, Xb15 band
S, its own band
0
2
4
6
8
10
t (ms)